

Hexagonal Architecture

by Samuel Runggaldier & Peter Unterthurner

ОТБРАТНО

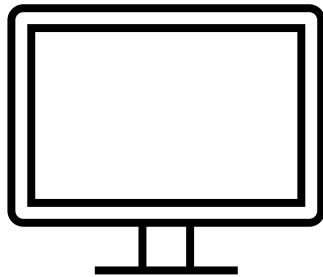
Content

1. How it came to be
2. Core concepts
3. Pro and Contra
4. How to implement it in an existing project
5. Code example

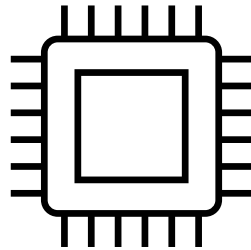
How it came to be

КОЛЛЕКЦИЯ

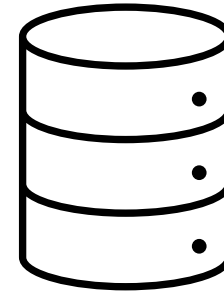
3 Tier Architecture



Presentation Tier

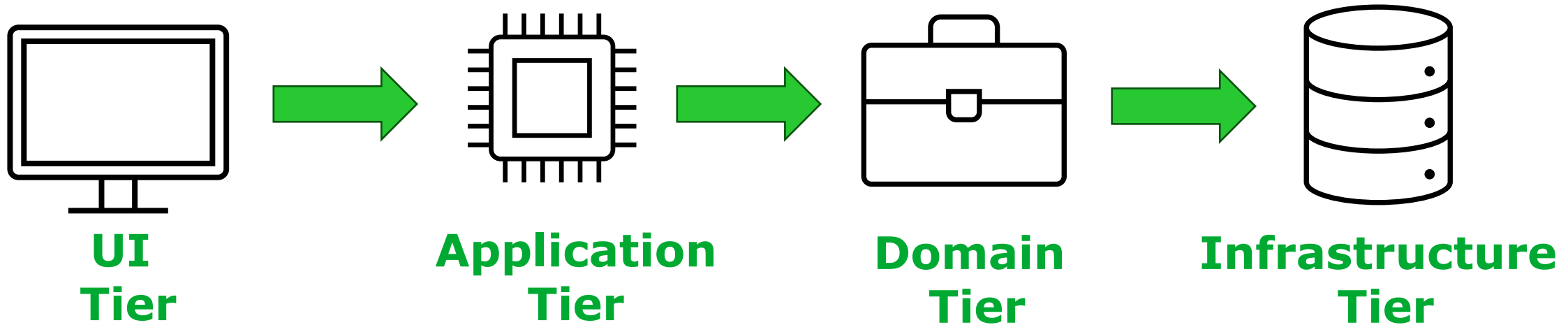


Logic Tier



Data Tier

Domain Driven Design: 4 Tier Architecture



Benefit of Layered Architecture

Benefit of Layered Architecture

Separation of concerns

Benefit of Layered Architecture

Separation of concerns

BUT

Benefit of Layered Architecture

Separation of concerns

BUT

No built-in mechanism to detect logic leaks between layers



Hexagonal Architecture

- Alistair Cockburn, 2005



Hexagonal Architecture

- Alistair Cockburn, 2005
- UI and databases are interchangeable external actors



Hexagonal Architecture

- Alistair Cockburn, 2005
- UI and databases are interchangeable external actors
- Application agnostic to the outside

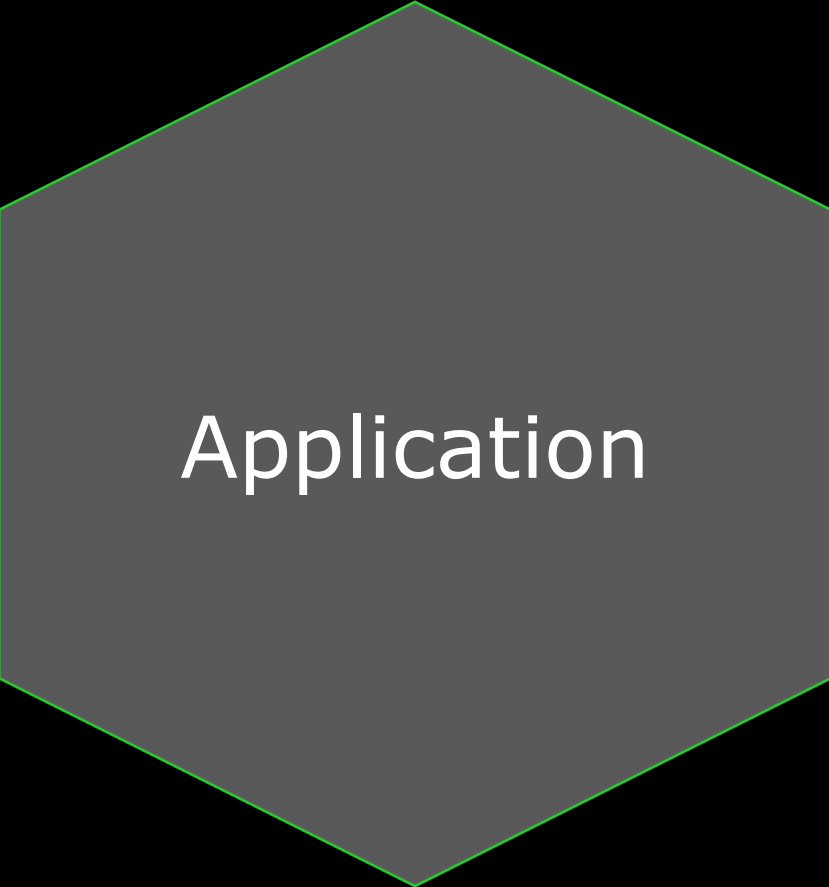


Hexagonal Architecture

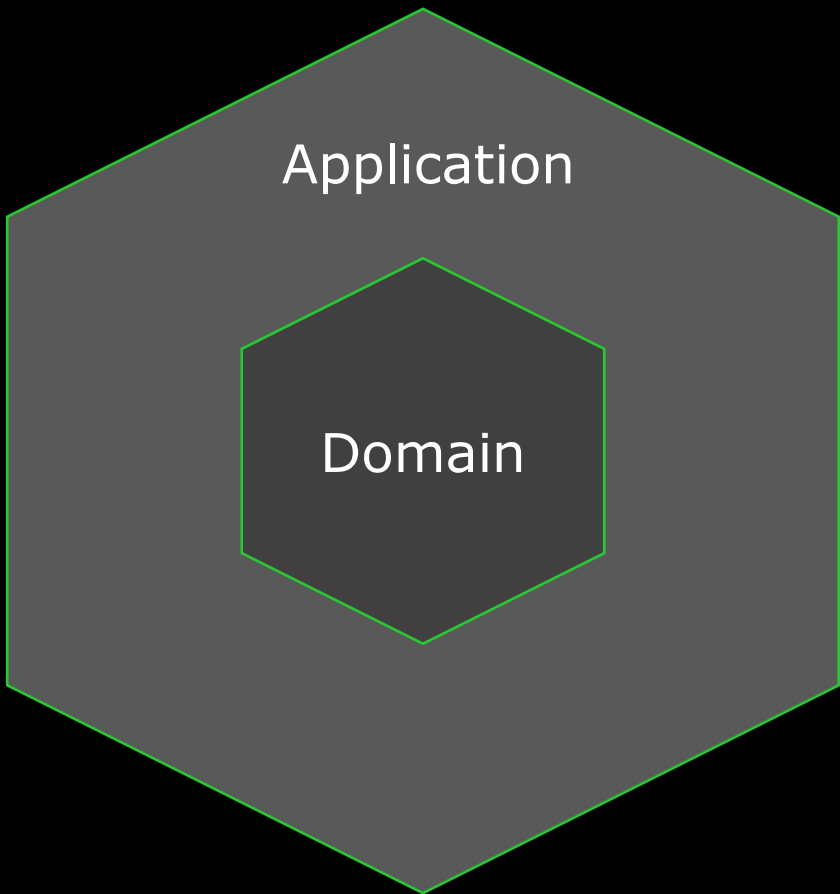
- Alistair Cockburn, 2005
- UI and databases are interchangeable external actors
- Application agnostic to the outside
- Ports & Adapters

Core Concepts

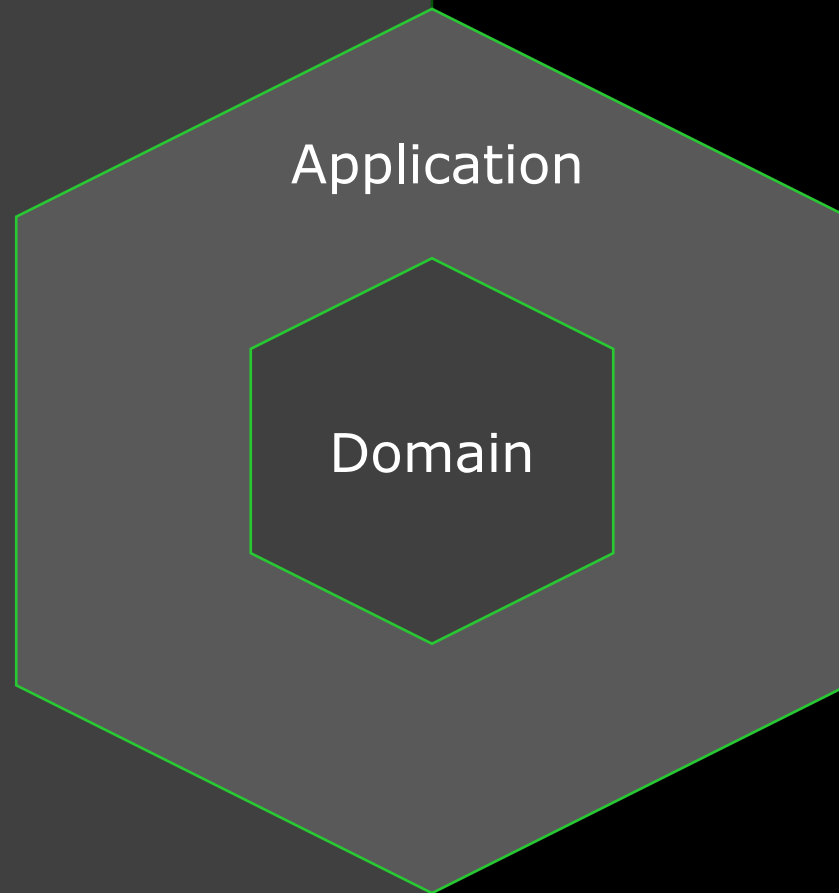
ОЛБРЕАНОХ



Application

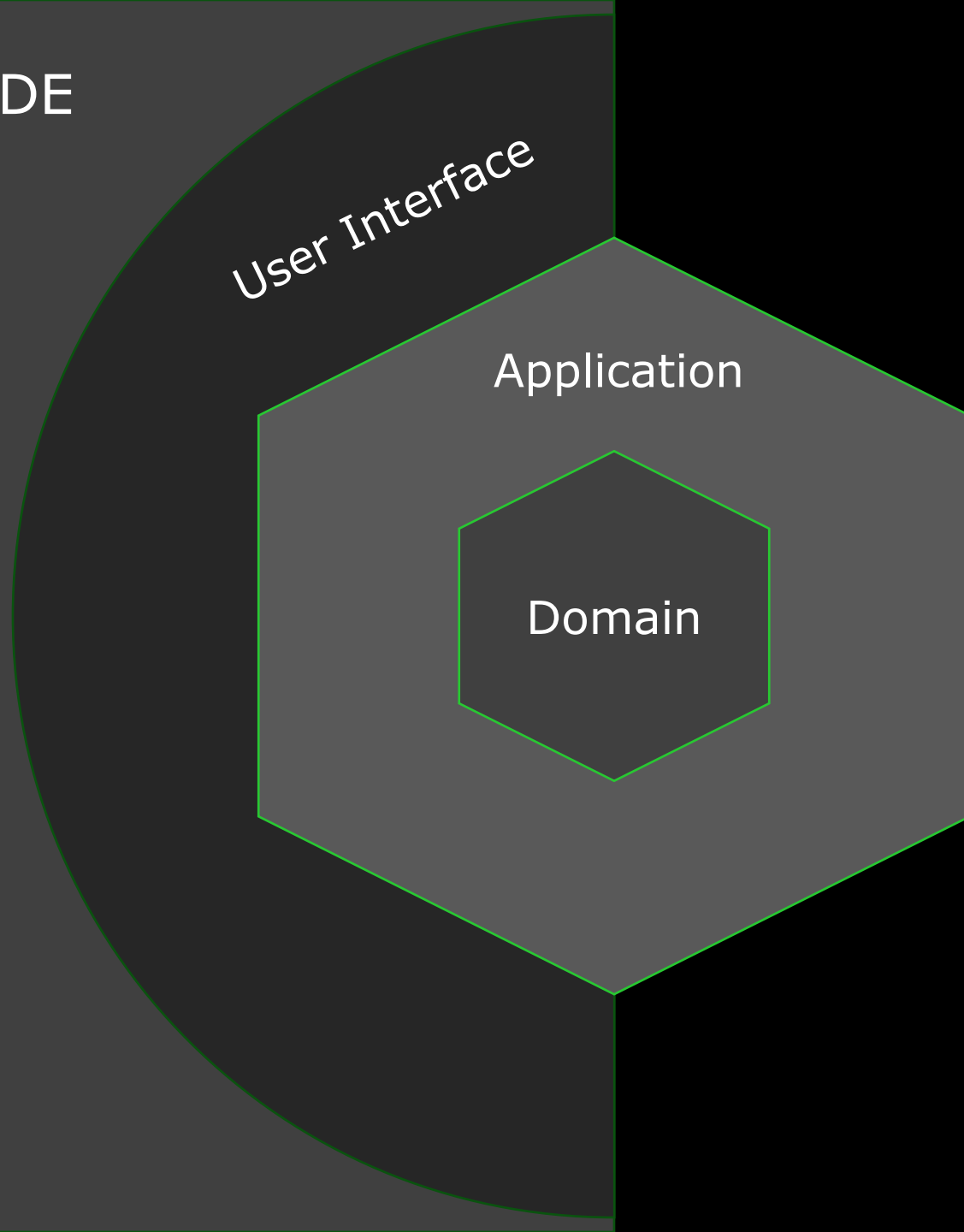


DRIVING SIDE



ΚΟΝΤΕΙΝΟΡ

DRIVING SIDE



КОЛЕСА

DRIVING SIDE

1010
1010



User Interface

Application

Domain

ОТБРАЗОК

DRIVING SIDE

DRIVEN SIDE

OLBREVNOY

1010
1010



User Interface

Application

Domain

DRIVING SIDE

DRIVEN SIDE

OLBRECHT

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1010



User Interface

Infrastructure

Application

Domain

DRIVING SIDE

DRIVEN SIDE

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User Interface

Infrastructure

Application

Domain



DRIVING SIDE

DRIVEN SIDE

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User Interface

Infrastructure

Application

Domain

Port

Port



Ports





Ports

- Technology-agnostic



Ports

- Technology-agnostic
- Defines interface for communication with the application



Ports

- Technology-agnostic
- Defines interface for communication with the application
- Dependency Inversion



Ports

- Technology-agnostic
- Defines interface for communication with the application
- Dependency Inversion
- **Driving Ports** (Inbound) & **Driven Ports** (Outbound)

Driving Ports: UseCase



Driving Ports: UseCase

- One Objective





Driving Ports: UseCase

- One Objective
- One Method

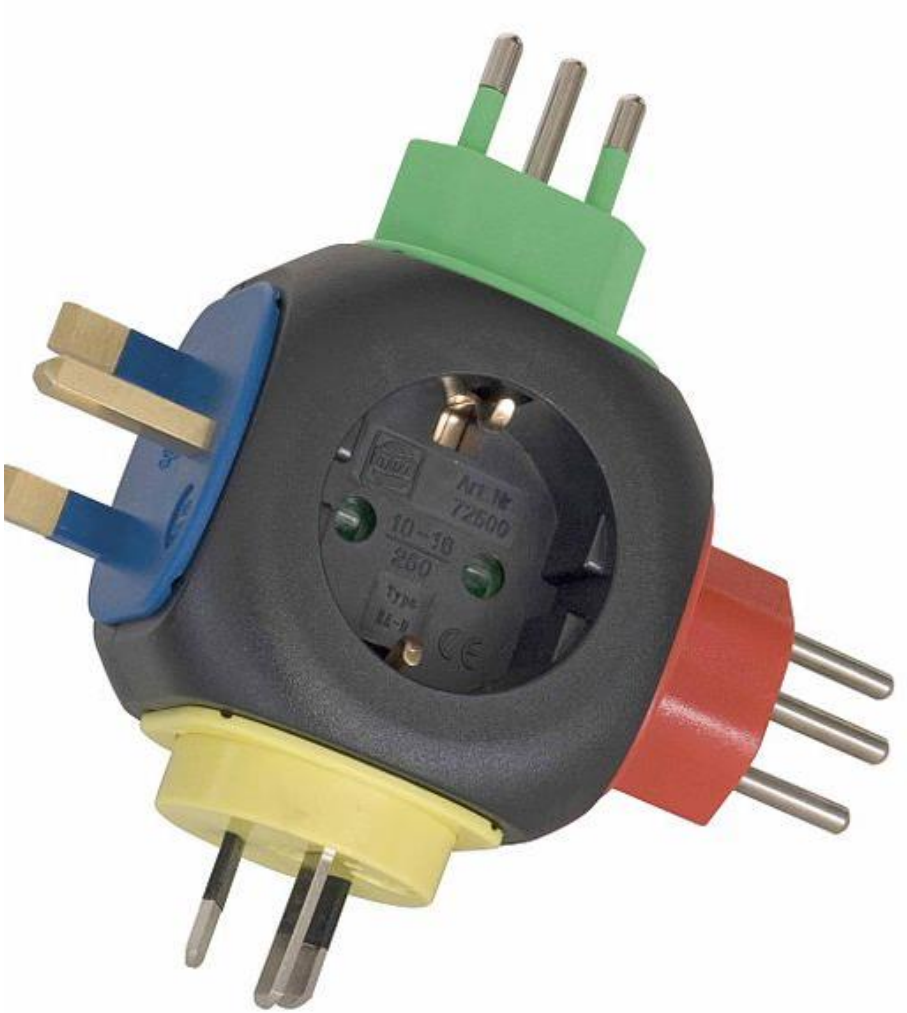


Driving Ports: UseCase

- One Objective
- One Method
- One Actor

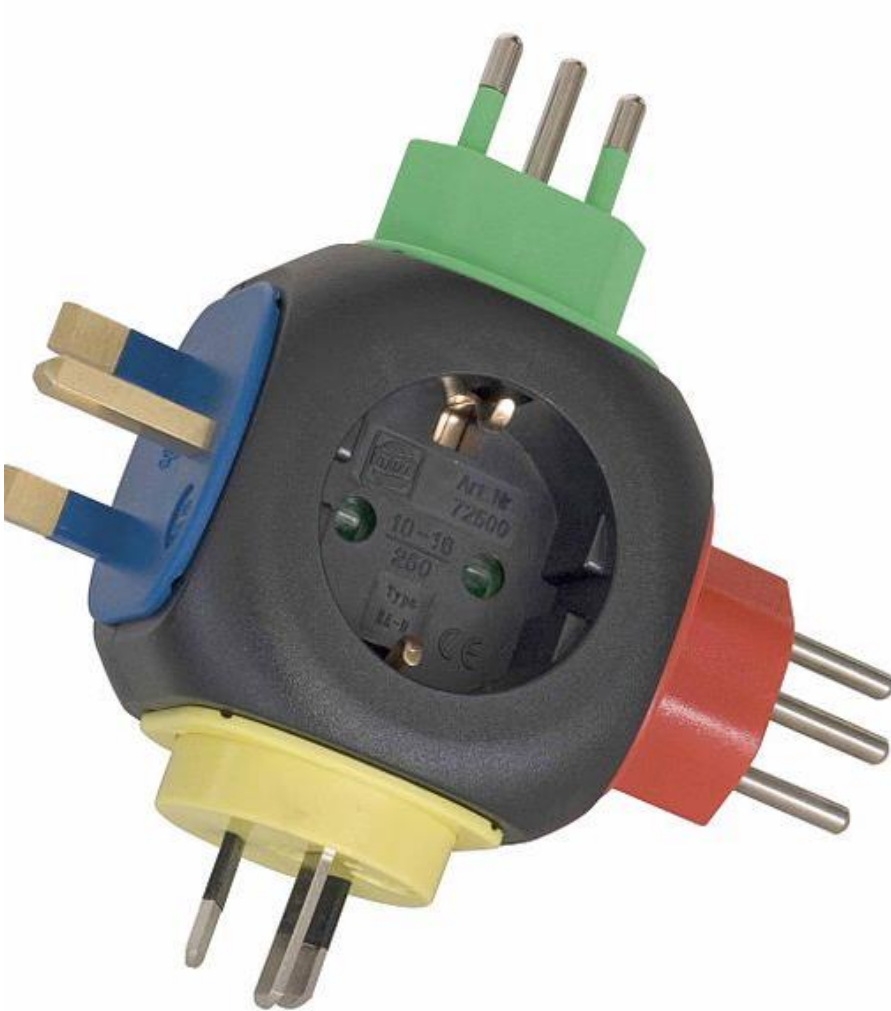


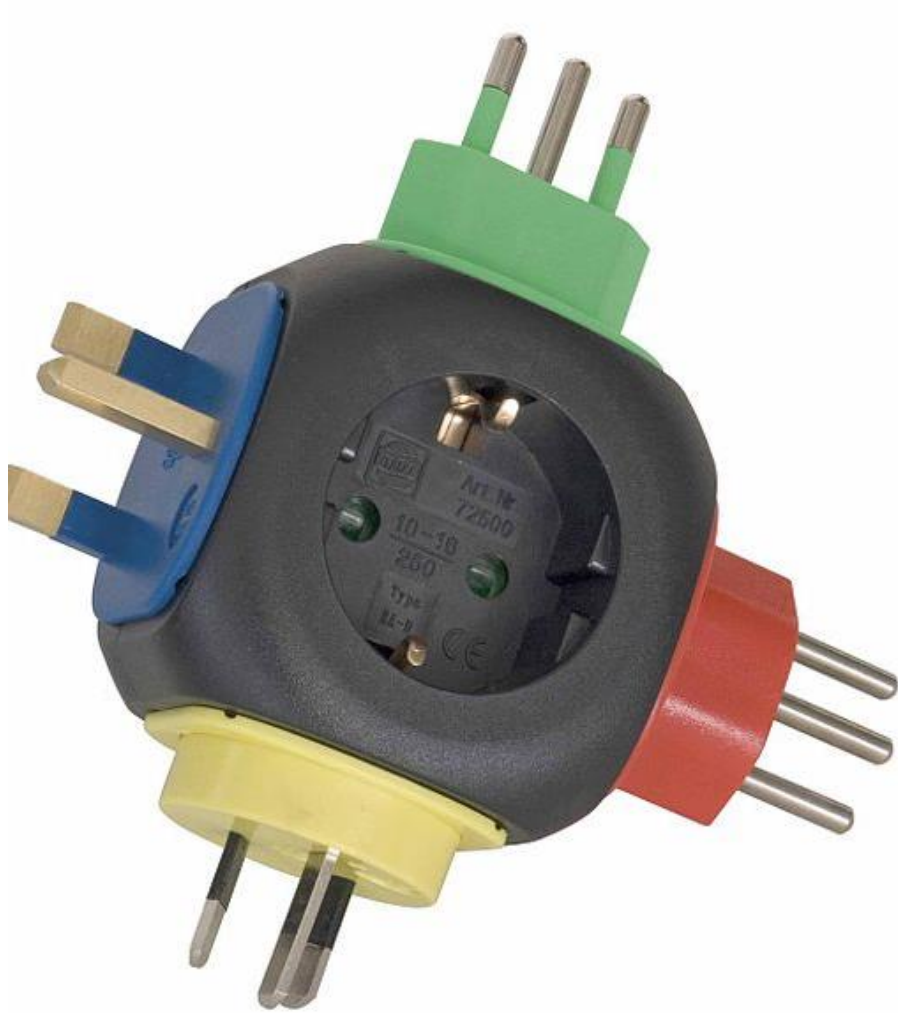
Adapter



Adapter

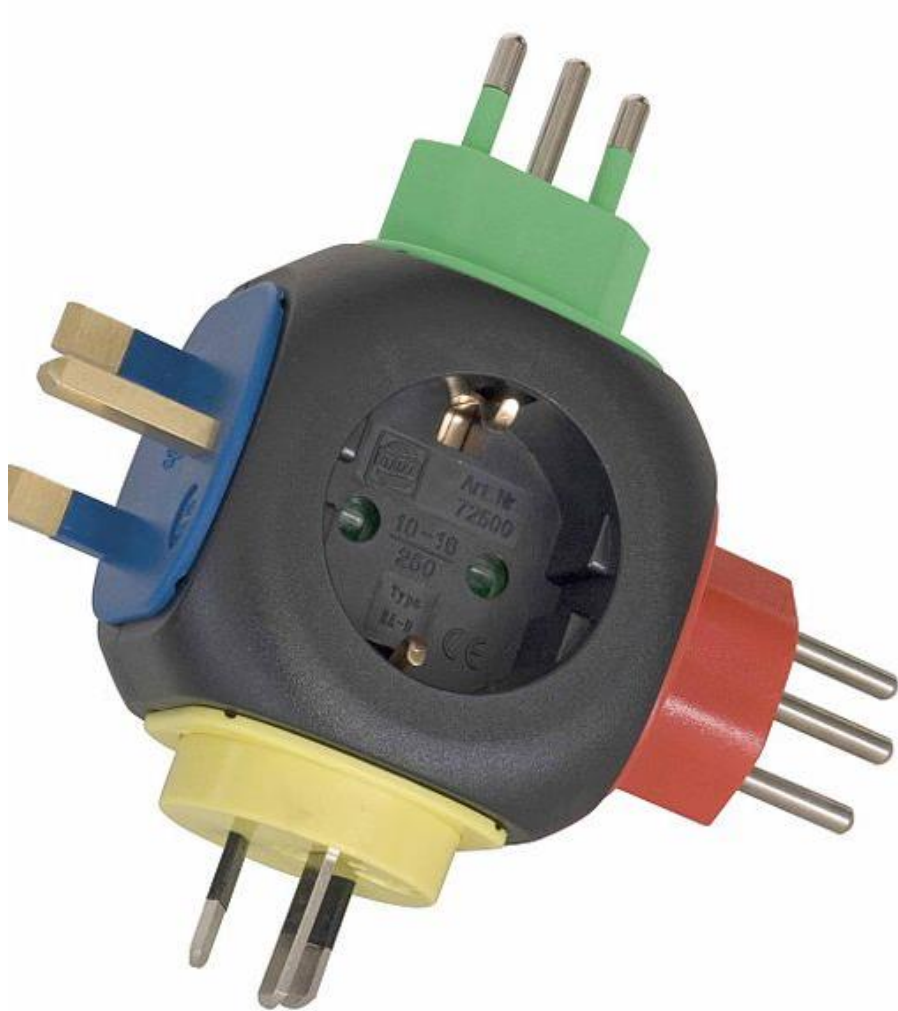
- Concrete details





Adapter

- Concrete details
- Conversion Layer



Adapter

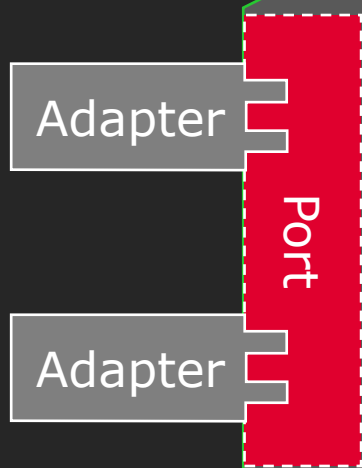
- Concrete details
- Conversion Layer
- Exchangable

DRIVING SIDE

DRIVEN SIDE

OLBREANNOY

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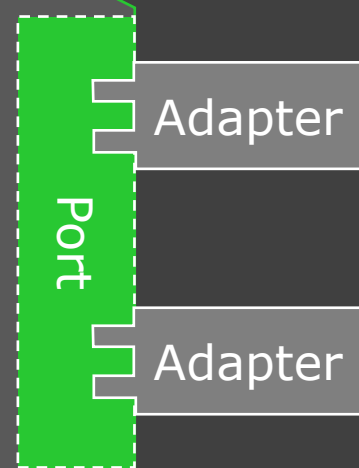


User Interface

Application

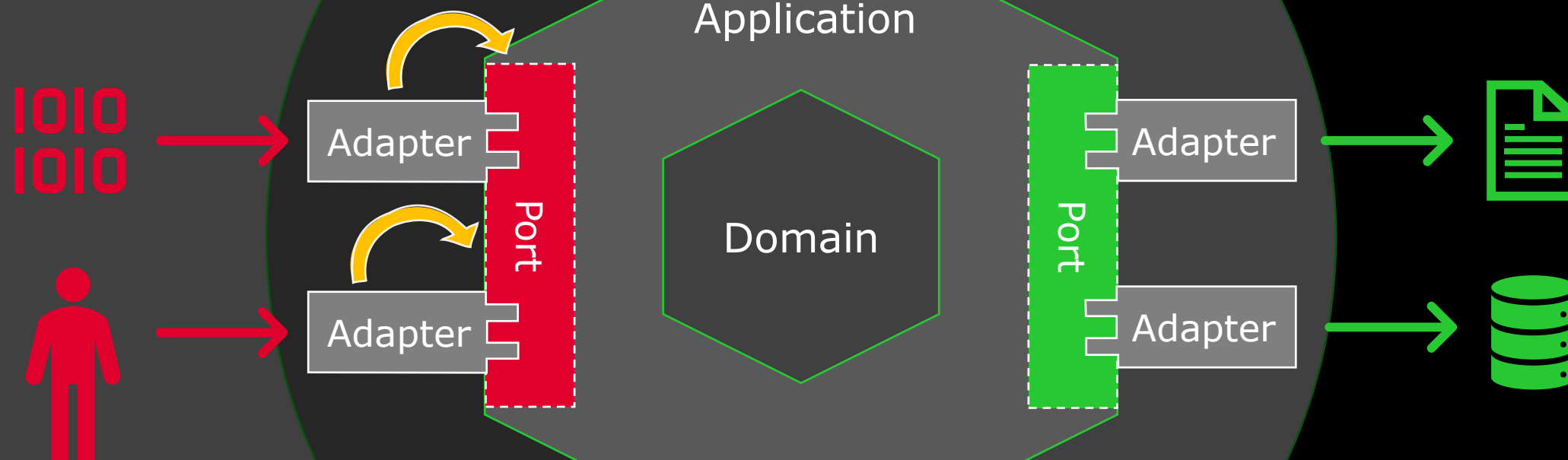
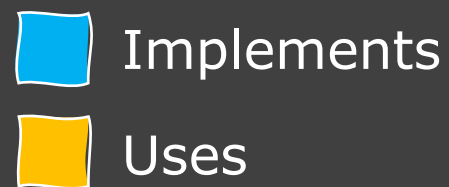
Domain

Infrastructure



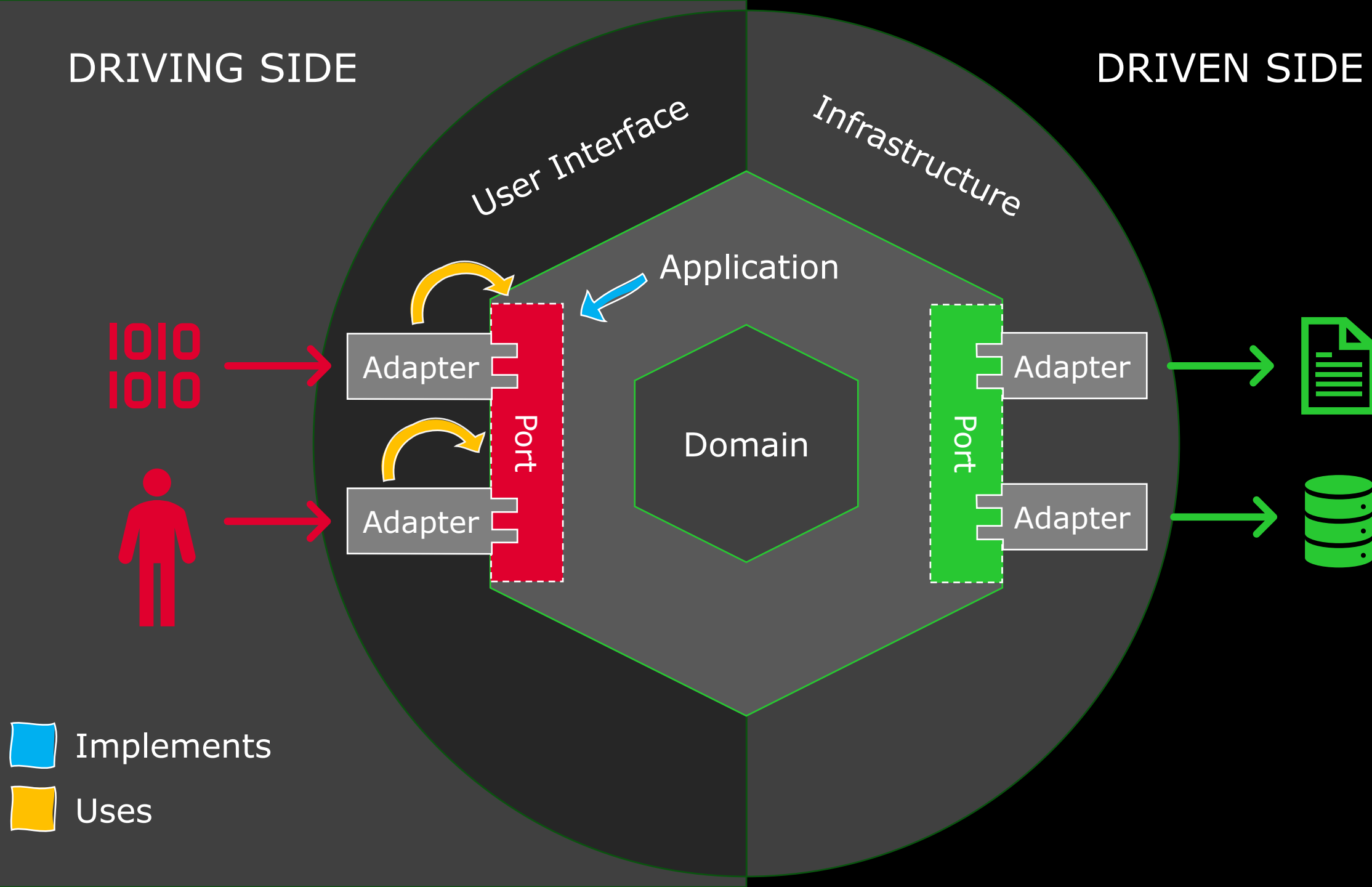
DRIVING SIDE

DRIVEN SIDE



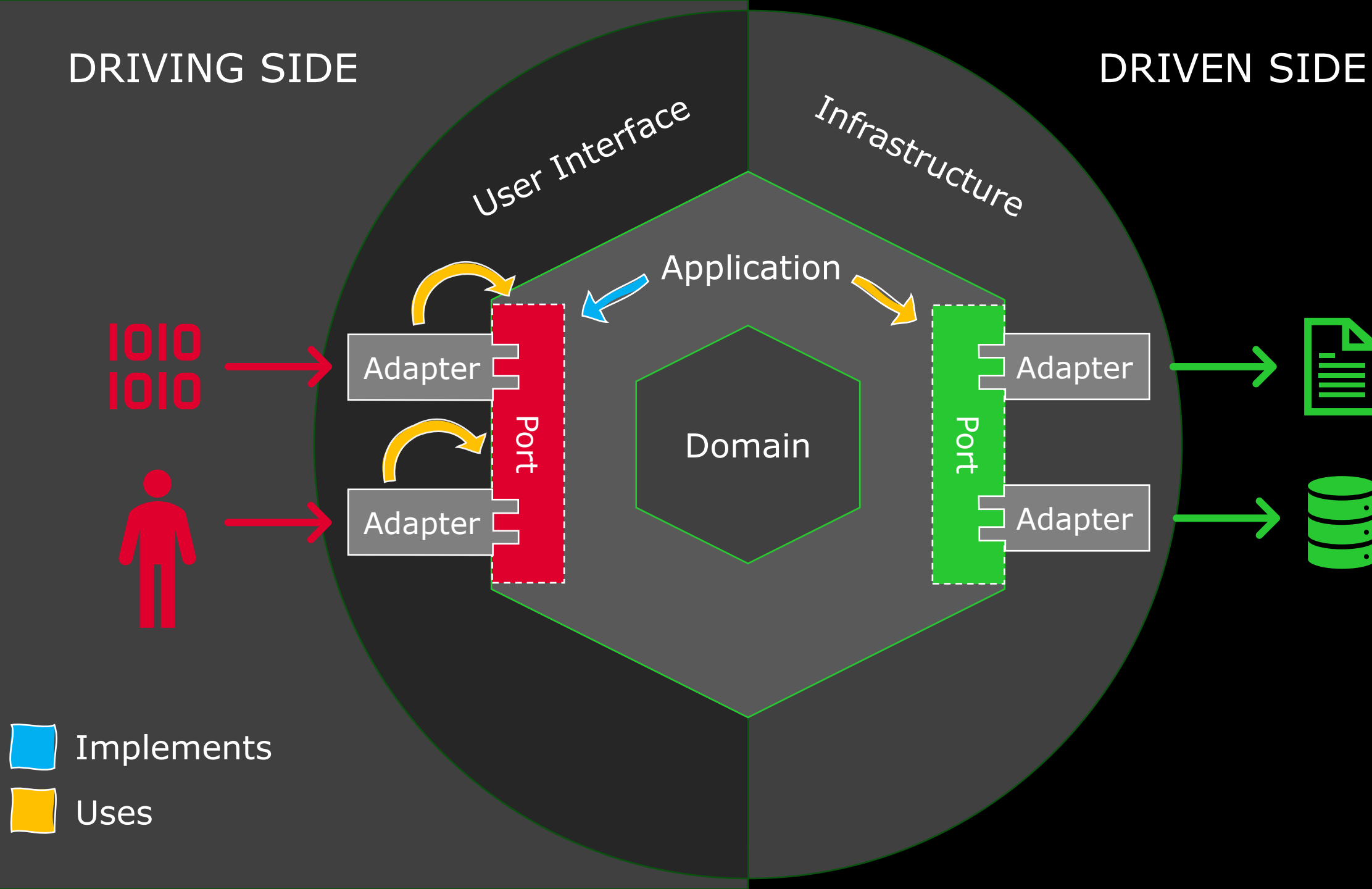
DRIVING SIDE

DRIVEN SIDE



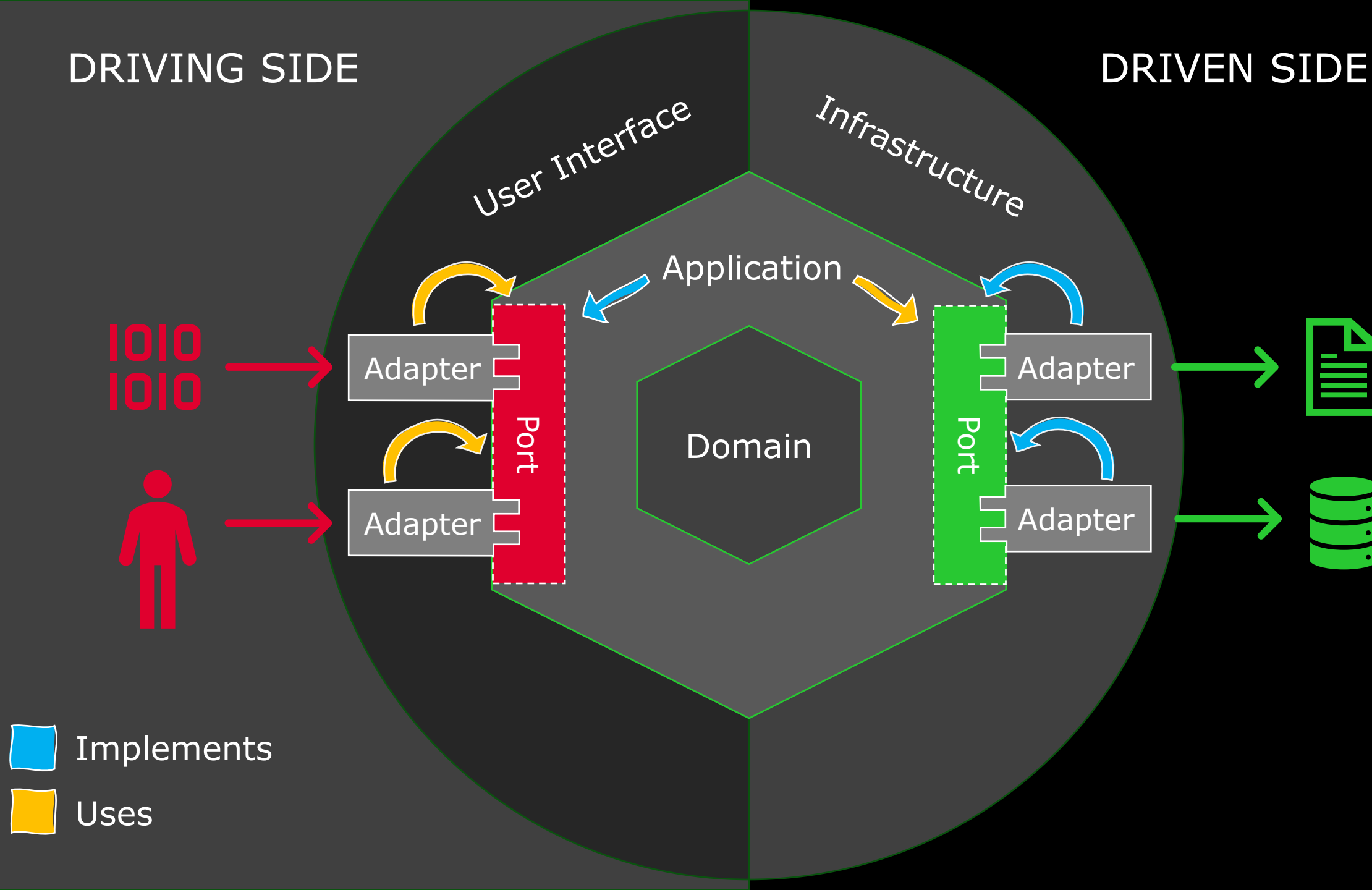
DRIVING SIDE

DRIVEN SIDE



DRIVING SIDE

DRIVEN SIDE

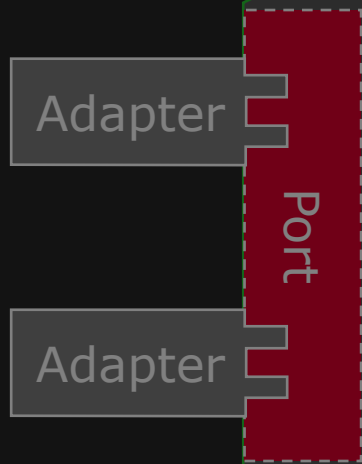
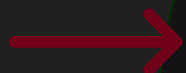


DRIVING SIDE

DRIVEN SIDE

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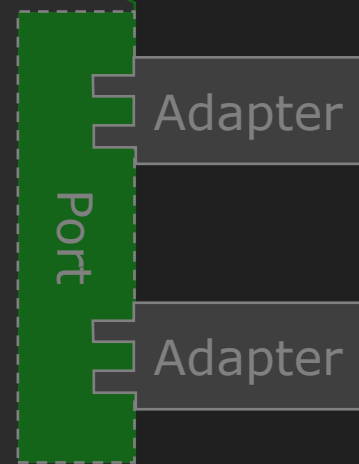


User Interface

Infrastructure

Application

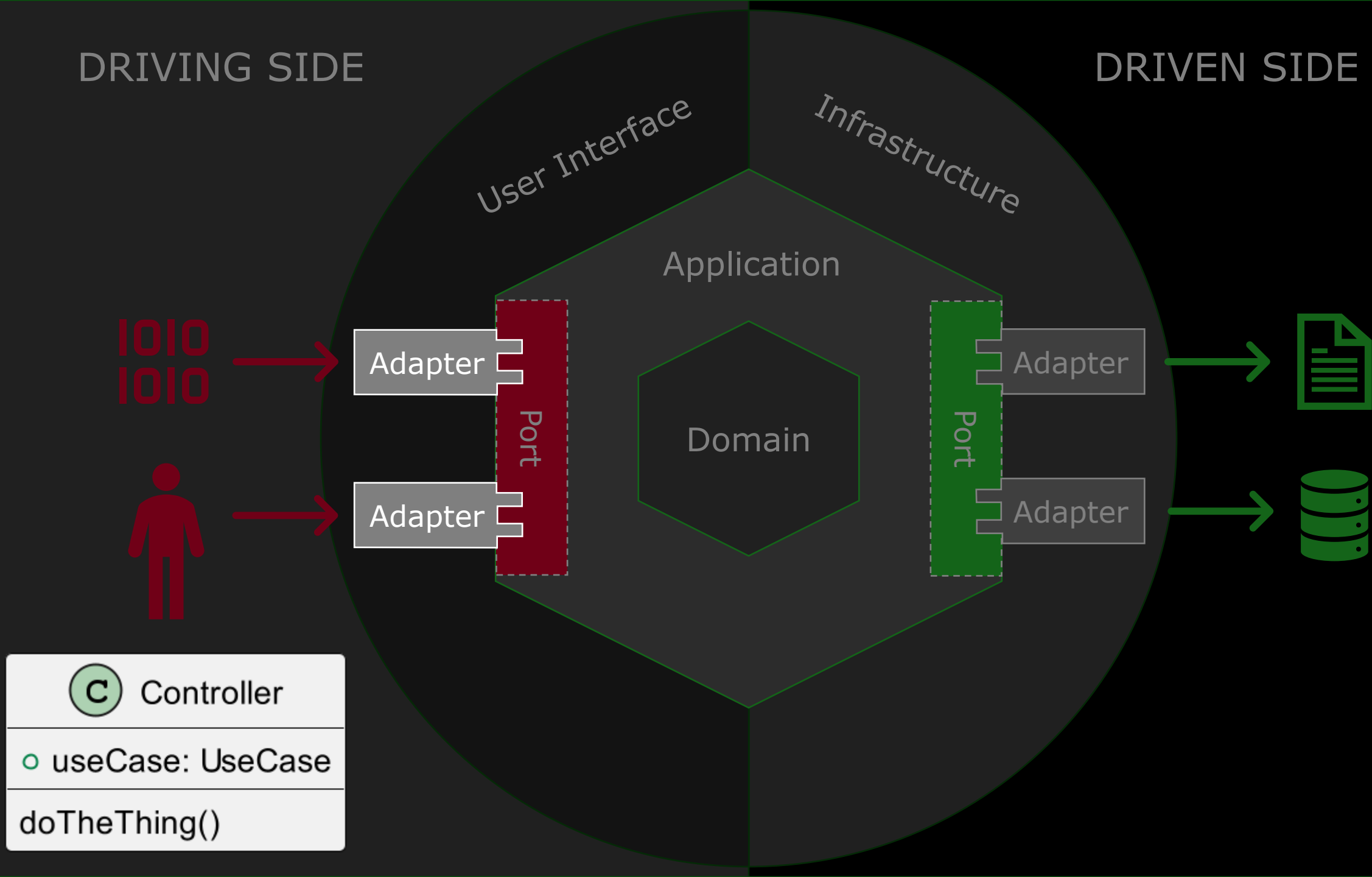
Domain



DRIVING SIDE

DRIVEN SIDE

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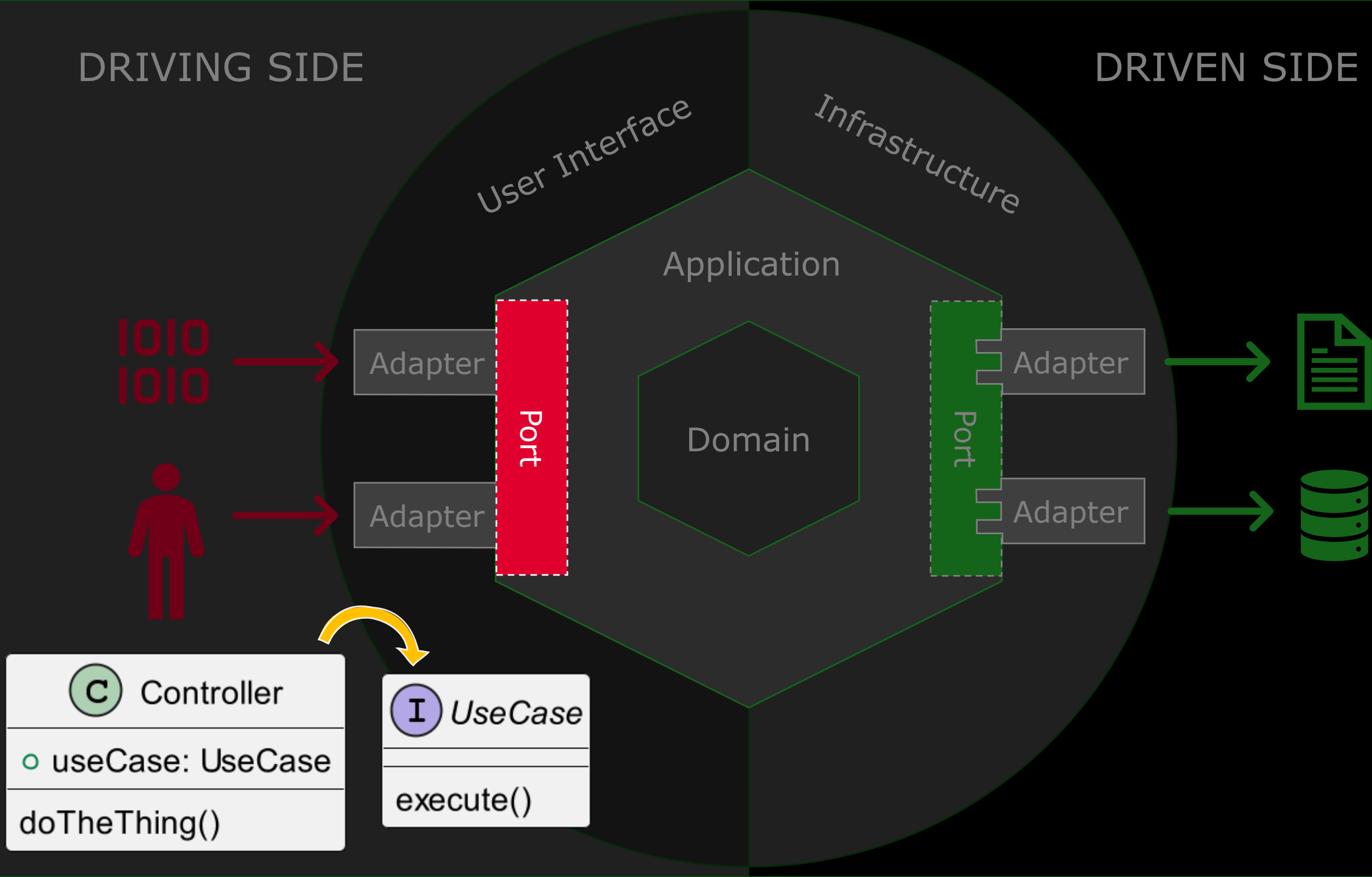
Controller

useCase: UseCase

doTheThing()

DRIVING SIDE

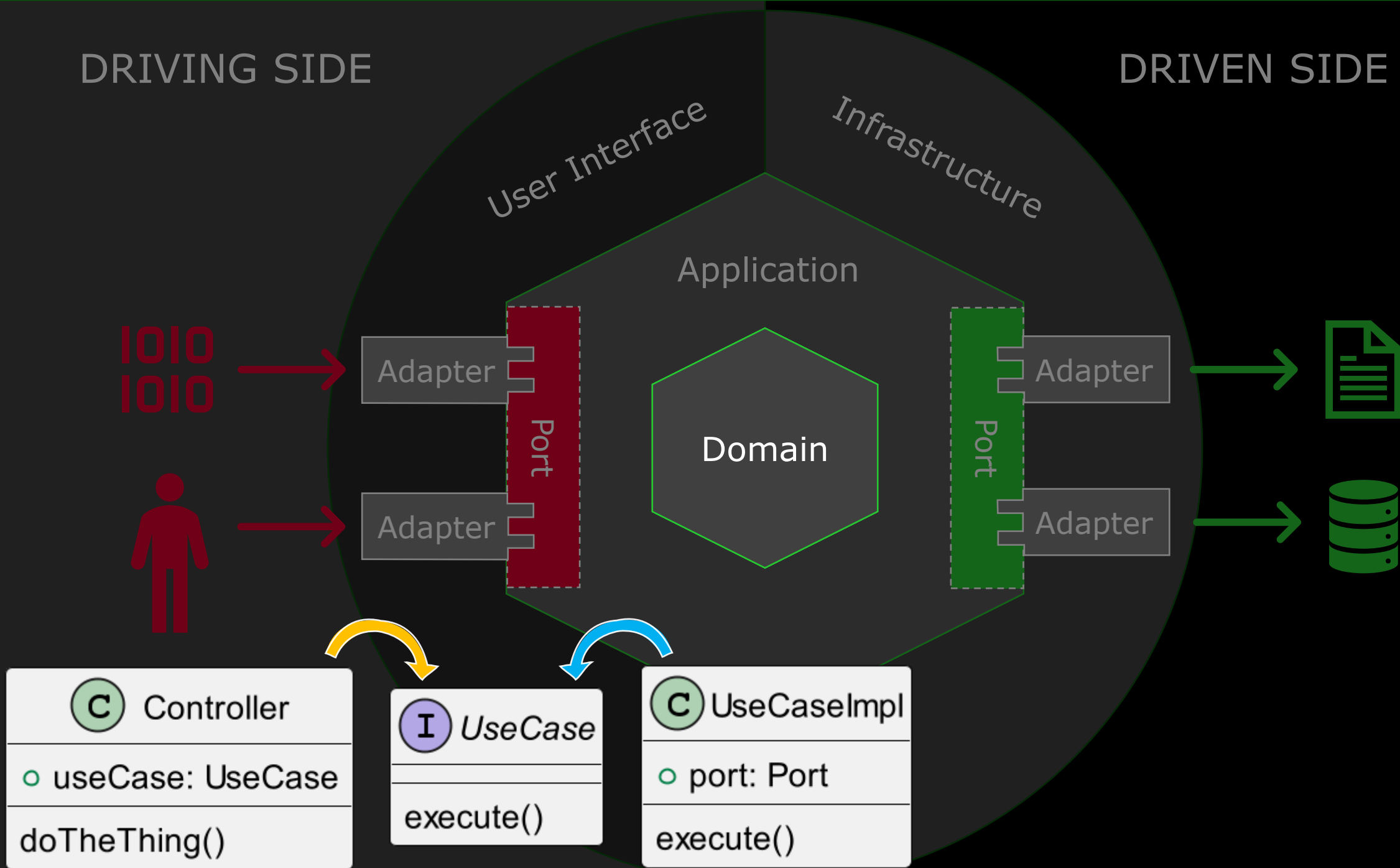
DRIVEN SIDE



DRIVING SIDE

DRIVEN SIDE

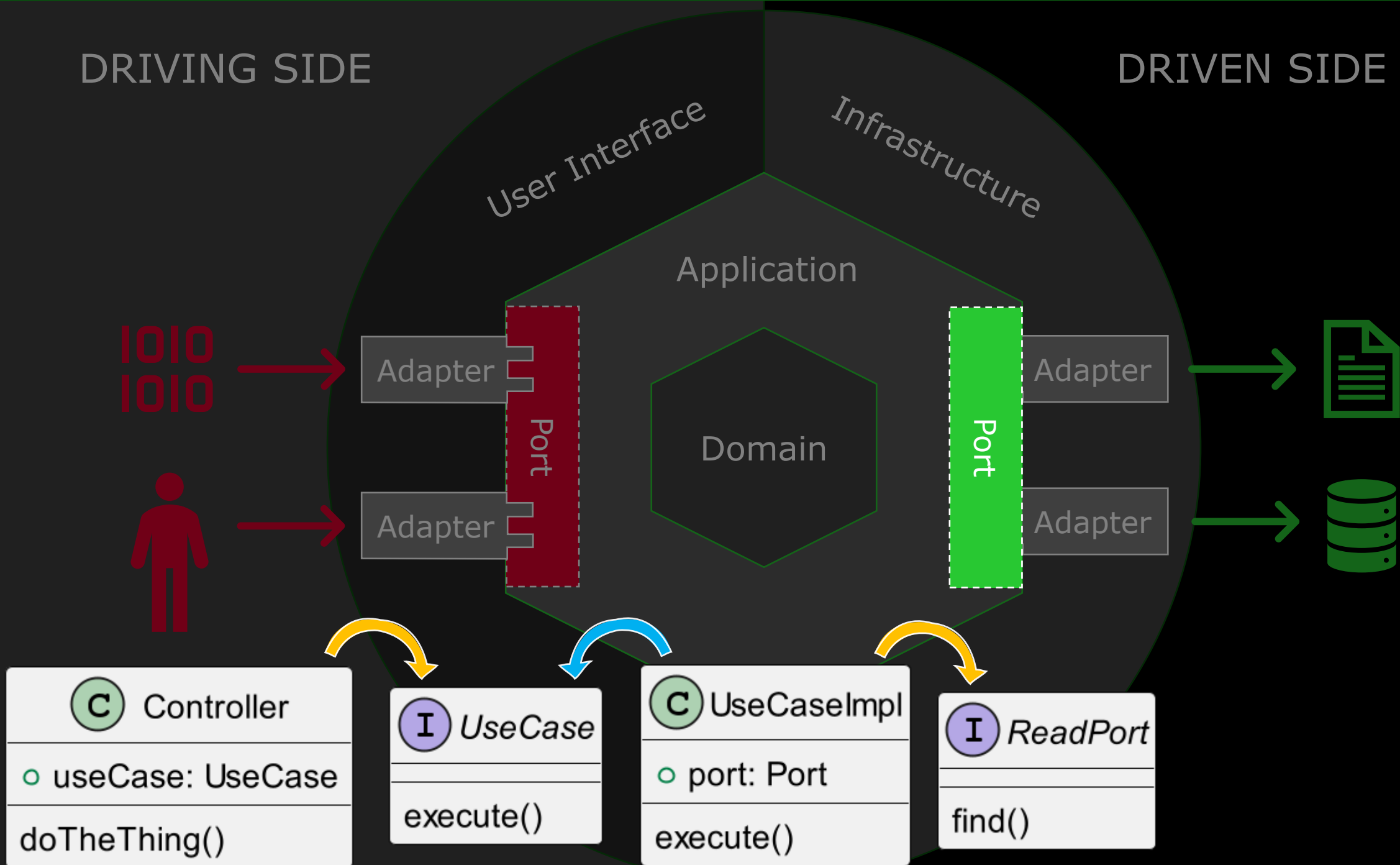
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DRIVING SIDE

DRIVEN SIDE

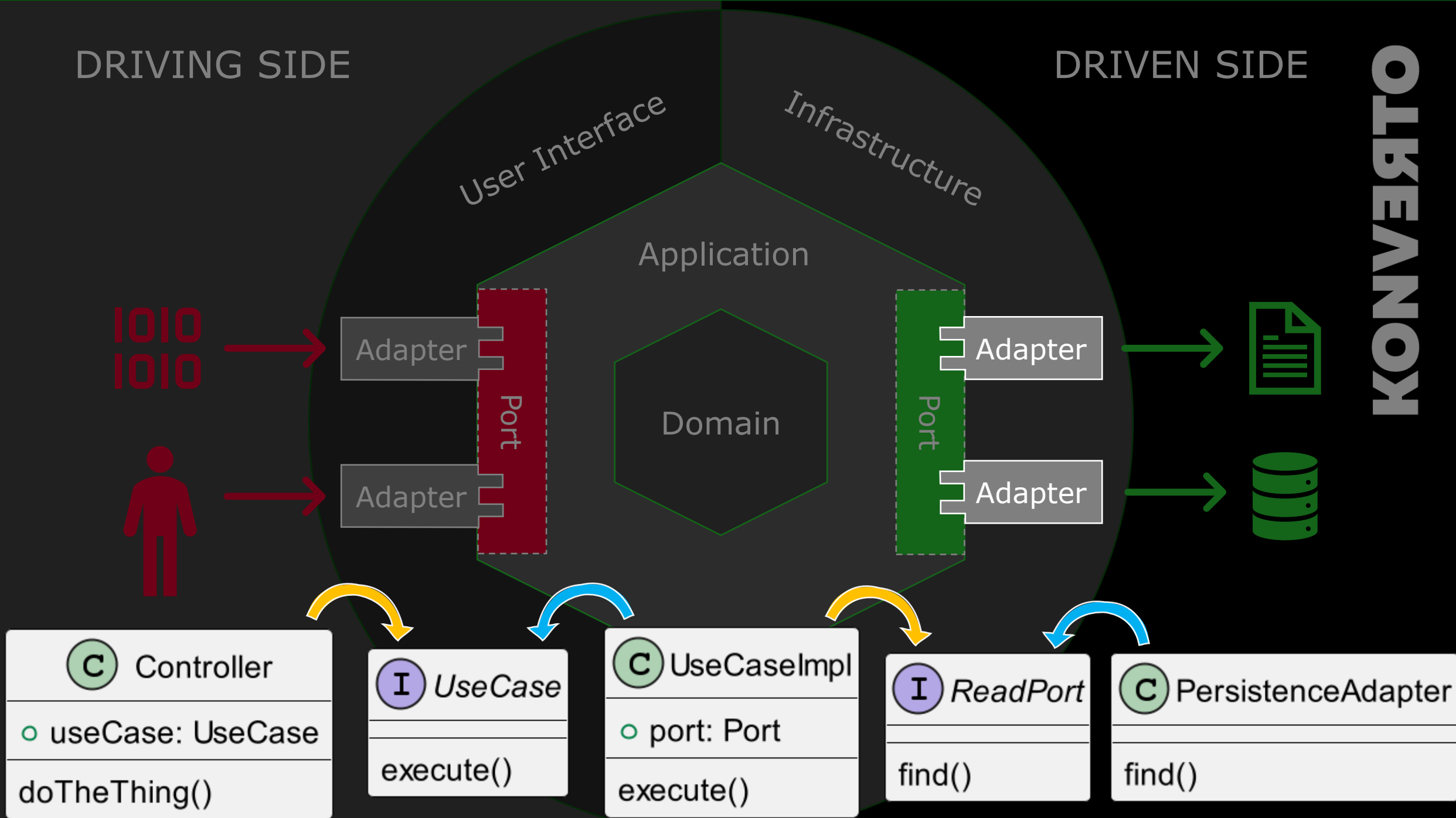
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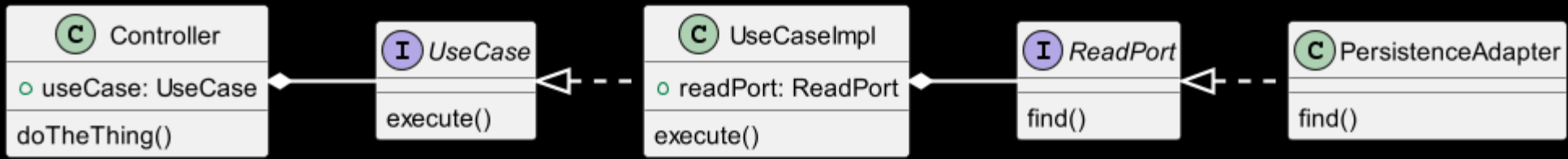


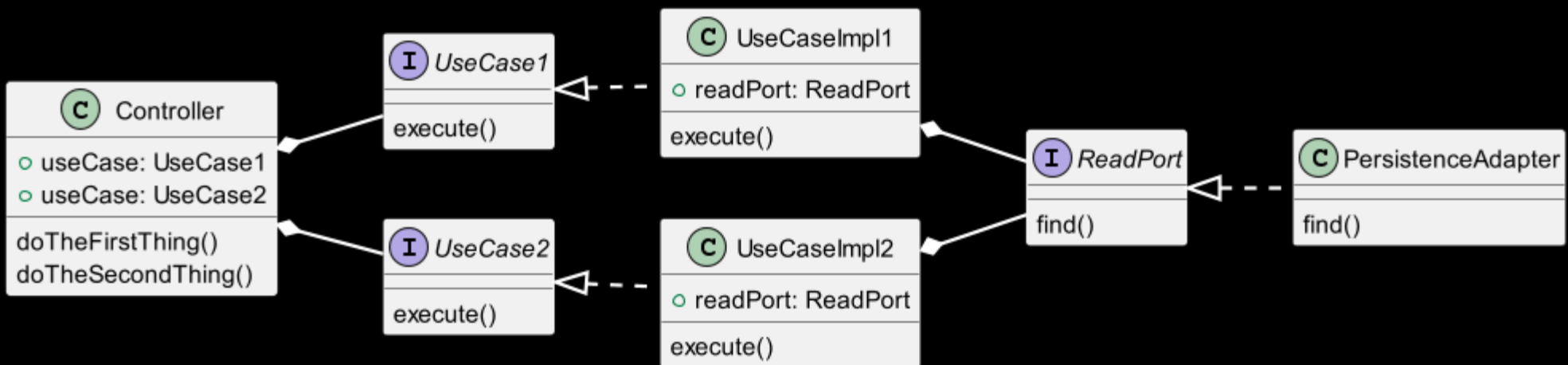
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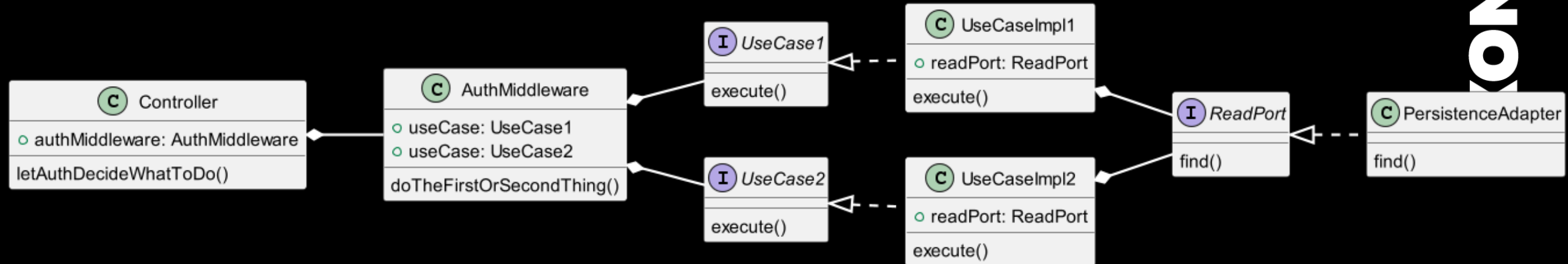
DRIVEN SIDE

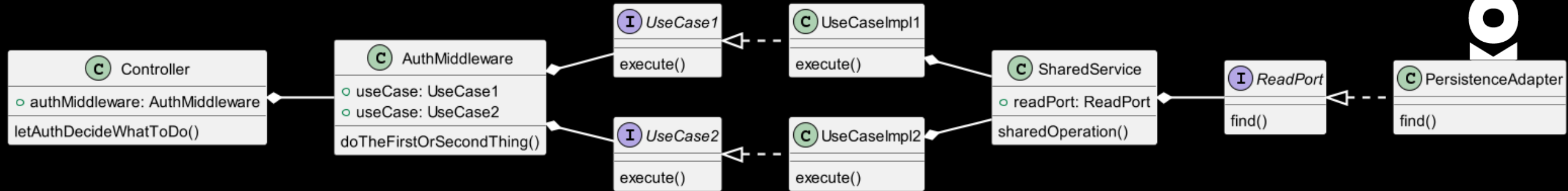
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Pro and Contra

ОЛБРЕАНОХ

Pro

Pro

- Scalable

Pro

- Scalable
- Flexible

Pro

- Scalable
- Flexible
- Testable

Contra

Contra

- Learning Curve

Contra

- Learning Curve
- Increased Complexity

Contra

- Learning Curve
- Increased Complexity
- Bigger Overhead for small projects

BUT

BUT

Will the project stay small?

BUT

Will the project stay small?

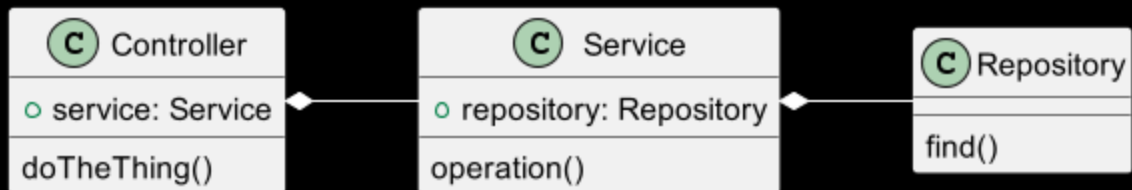
There will be no Jira Ticket „Refactoring Architecture“

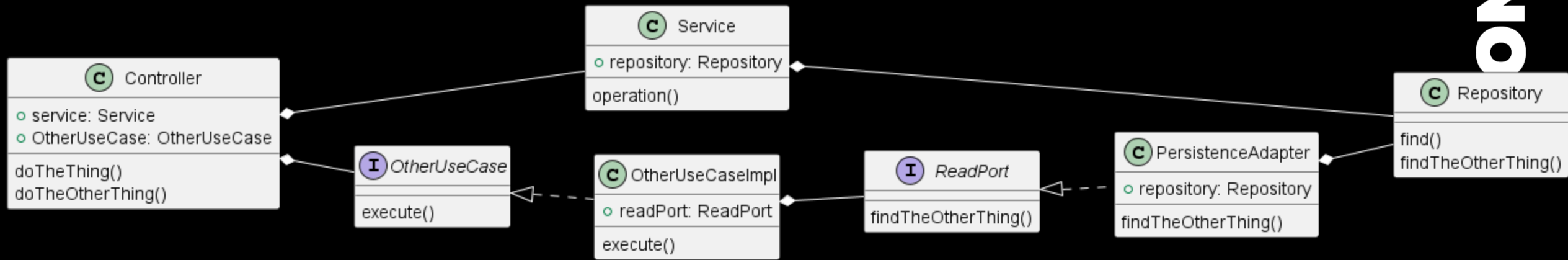
**How to implement it
in an existing
project?**

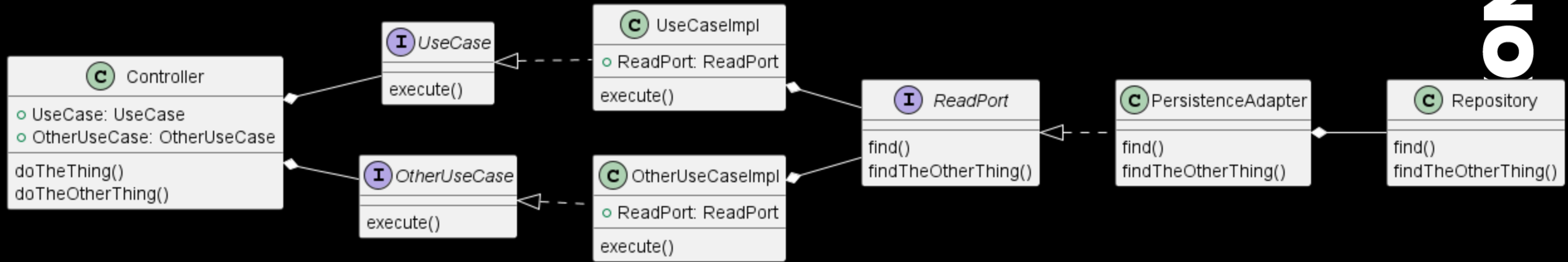
ОЛБРЕАНОВ



КОНВЕНЦИИ







Code Example

ОТБЕЛНОВ